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School of Computer Science and Artificial Intelligence

Lab Assignment # 10.2

Program	**: B. Tech (CSE)**
Specialization	**: -**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**: II**
Academic Session	**: 2025-2026**
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Enrollment No.	**: 2403A51L53**
Batch No	**: 52**
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Task -01(Error Detection and Correction)

Code used:

```
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
```

```
  File "/tmp/ipython-input-1993060875.py", line 1
    def calculate_total(nums)
                             ^
SyntaxError: expected ':'
```

Prompt: Correct the code and give brief explanations of the identified syntax and logic errors.

Output:

```
def calculate_total(nums):
    total_sum = 0
    for n in nums:
        total_sum += n
    return total_sum
```

Explaination:

1) Missing colons: Python requires a colon : at the end of def (function definition) and for (loop definition) statements.

2) Variable shadowing: I renamed the sum variable to total_sum to avoid overwriting Python's built-in sum() function, which can lead to unexpected behavior.

3) Undefined variable: The function was trying to return total, but the variable was named sum (or total_sum after the fix). I've corrected it to return total_sum.

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Task-02(Code Style Standardization)

Code Used:

```
def findSum(a,b):return a+b
print(findSum(5,10))
```

```
15
```

Prompt: convert the code to follow standard style conventions.

Output:

```
def find_sum(a, b):
    return a + b

print(find_sum(5, 10))
```

```
15
```

Task-03(Code Clarity Improvement)

Code used:

```
def f(x,y):
return x-y*2
print(f(10,3))
```

```
  File "/tmp/ipython-input-2982719317.py", line 2
    return x-y*2
    ^
IndentationError: expected an indented block after function definition on line 1
```

Prompt: rewrite the code with meaningful function and variable names, proper indentation, and improved clarity.

Output:

```
def calculate_value(first_number, second_number):
    """Calculates the first number minus two times the second number."""
    result = first_number - (second_number * 2)
    return result

# Example usage with meaningful variable names for the call
input_a = 10
input_b = 3
output = calculate_value(input_a, input_b)
print(output)
```

```
4
```

Task-04(Structural Refactoring)

Code used:

```
print("Hello Ram")
print("Hello Sita")
print("Hello Ravi")

Hello Ram
Hello Sita
Hello Ravi
```

Prompt:make the code modular using reusable functions to eliminate repetition.

Output:

```
Gemini
def greet_person(name):
    print(f"Hello {name}")

greet_person("Ram")
greet_person("Sita")
greet_person("Ravi")

Hello Ram
Hello Sita
Hello Ravi
```

Task-05(Efficiency Enhancement)

Code used:

```
numbers = [ ]
for i in range(1, 500000):
  numbers.append(i * i)
print(len(numbers))

499999
```

Prompt: Optimize the code to achieve the same result with improved performance.

Output:

```
Gemini
numbers = [i * i for i in range(1, 500000)]
print(len(numbers))

499999
```